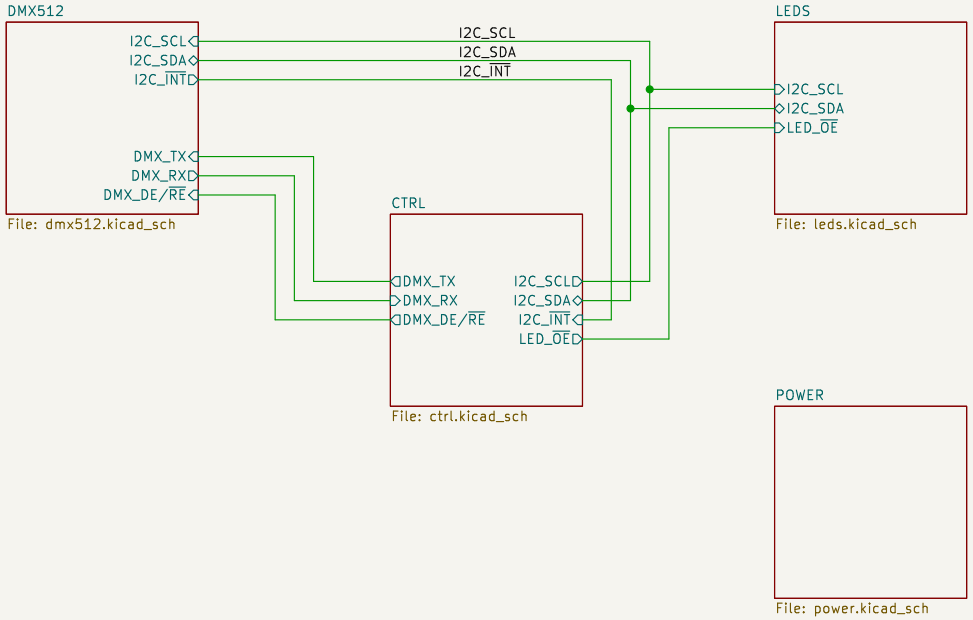
LED Beacon



H1 MountingHole

H2 MountingHole

H3 MountingHole

H4 MountingHole

Wires & Bits

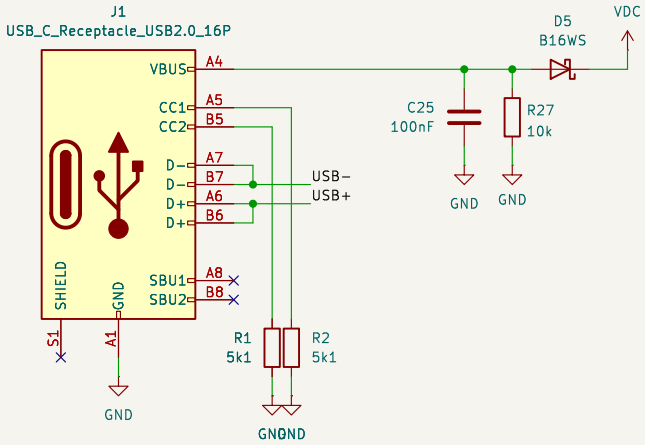
Sheet: /
File: emf_beacon.kicad_sch

Title: LED Beacon

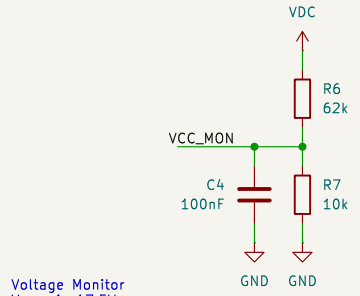
Size: A4	Date: 2026-06-05	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 1/5

Controller

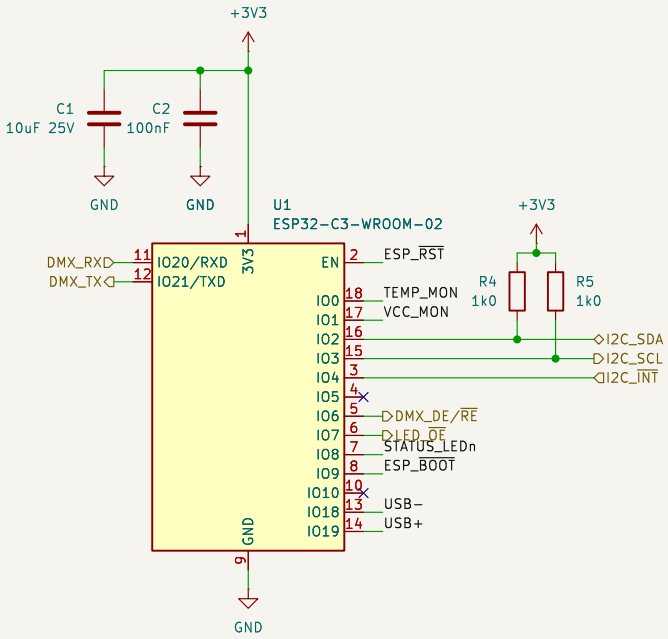
USB Programming



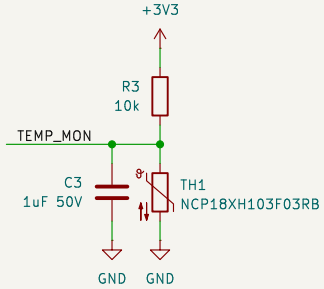
VCC Monitoring



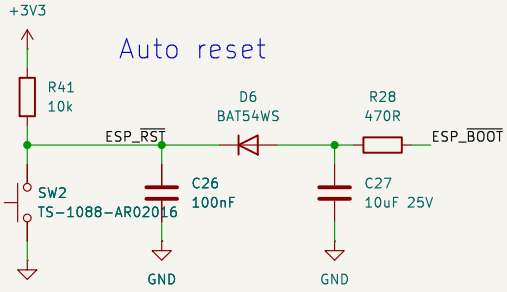
Voltage Monitor
Vmax is 17.5V
Vadc × ((43kΩ + 10kΩ) ÷ 10kΩ) = VCC
Vadc × 5.3 = VCC
1lsb = 4,271µV



Temperature Monitoring

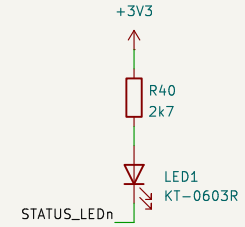


Auto reset



Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

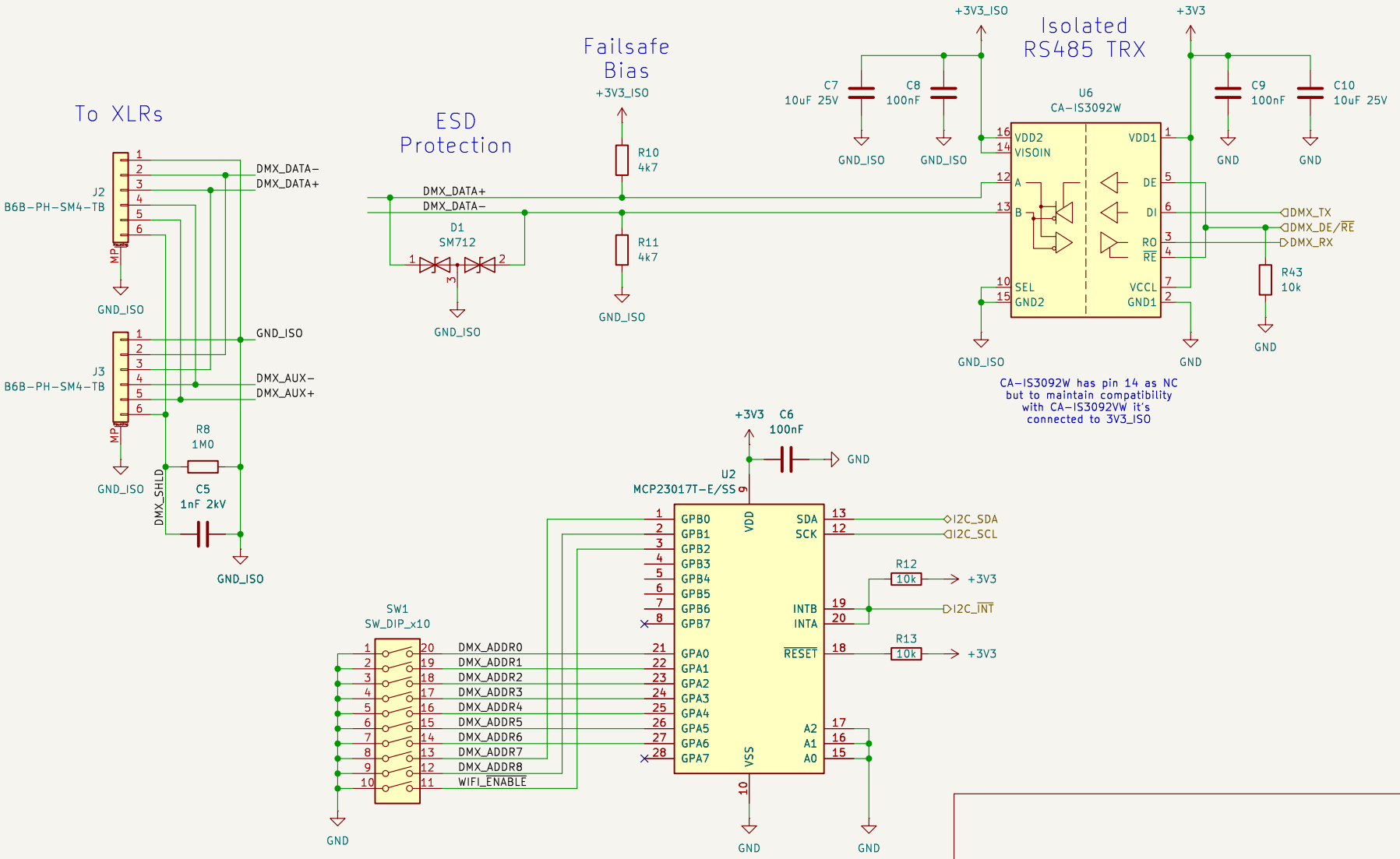
Status LED



IO8 needs to be held high to boot
so LED & resistor are high side.
Set STATUS_LEDn to 0V to light LED.

Wires & Bits		
Sheet: /CTRL/		
File: ctrl.kicad_sch		
Title: Controller		
Size: A4	Date: 2026-06-05	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 2/5

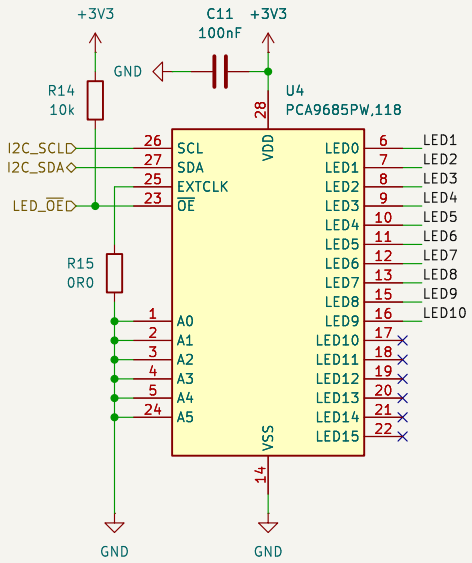
DMX512



Wires & Bits	
Sheet: /DMX512/	
File: dmx512.kicad_sch	
Title: DMX512	
Size: A4	Date: 2026-06-05
KiCad E.D.A. 9.0.9	Rev: 1
	Id: 3/5

LED Driver

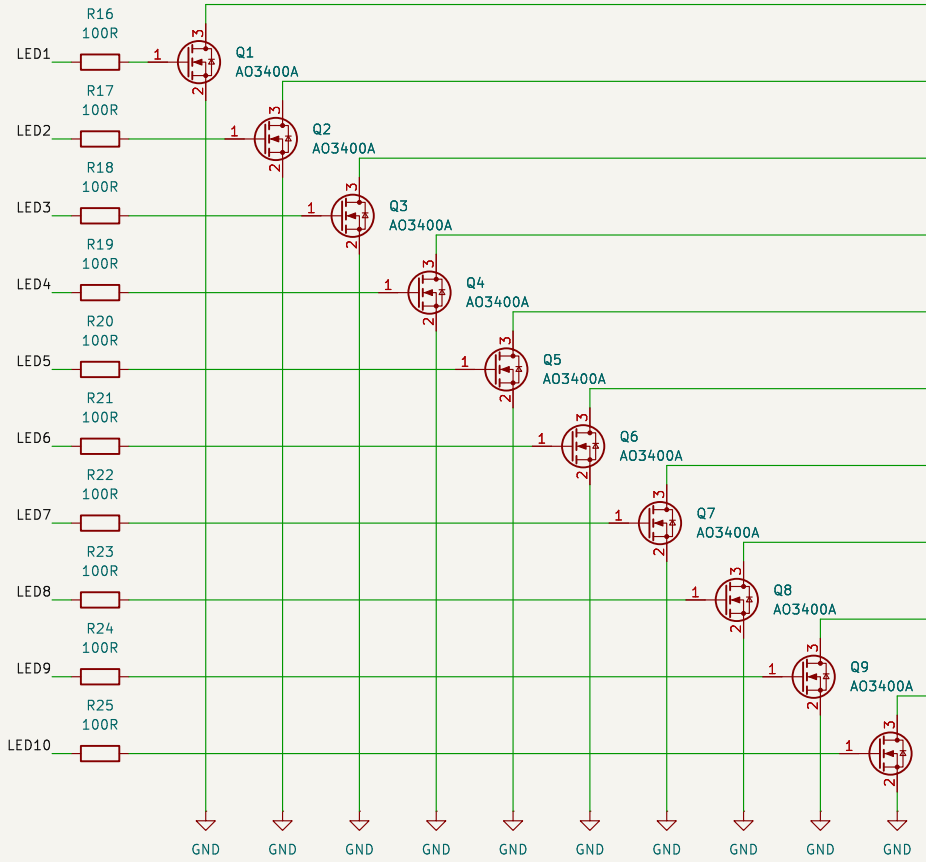
I2C → PWM



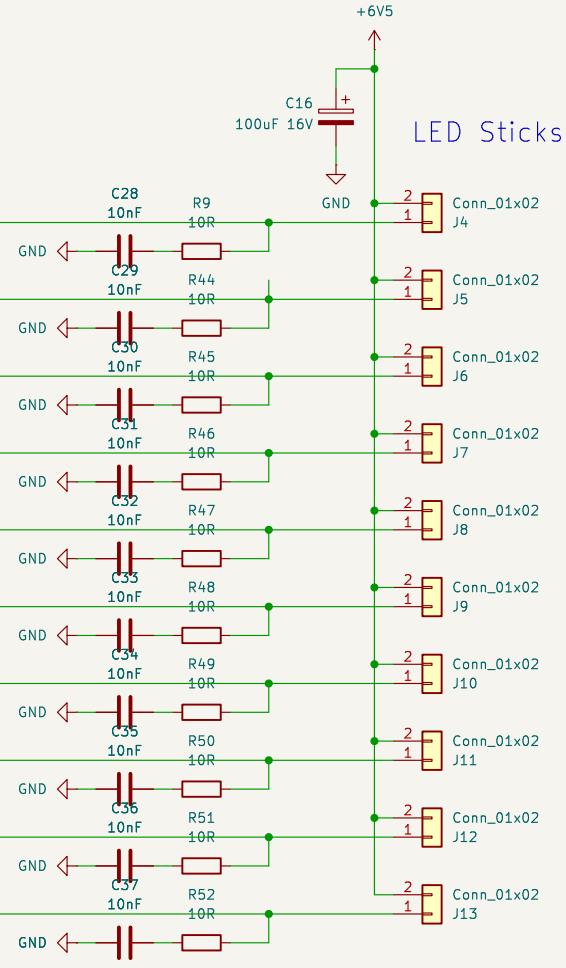
I2C Addr: 0x40

A string of 6x LEDs is 6.4Vf @ 0.3A

LED Enable



LED Sticks



Wires & Bits

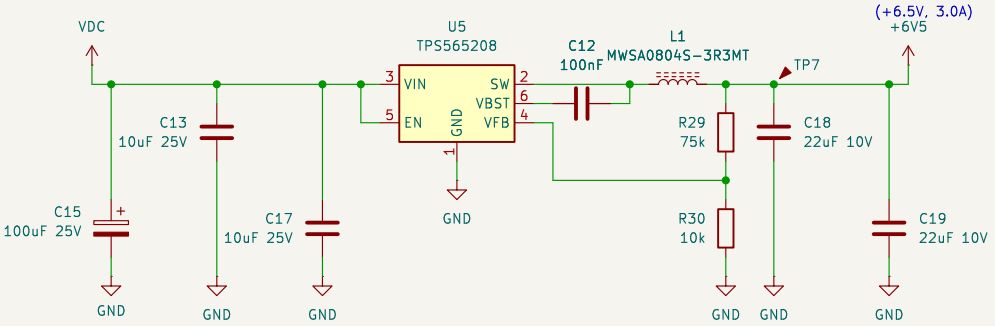
Sheet: /LEDS/
File: leds.kicad_sch

Title: LEDs

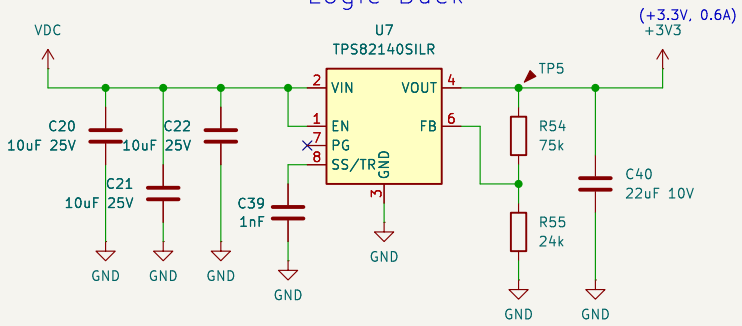
Size: A4	Date: 2026-06-05	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 4/5

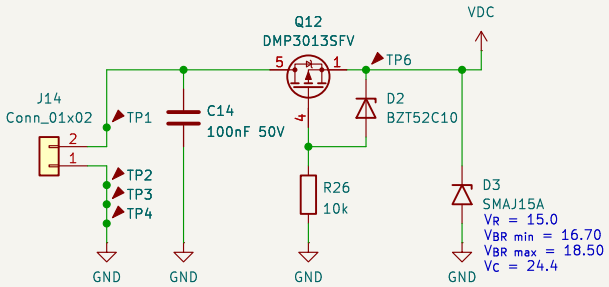
Power Supplies

LED Buck



Logic Buck





EMC ferrites /CMC?

Wires & Bits		
Sheet: /POWER/		
File: power.kicad_sch		
Title: Power		
Size: A4	Date: 2026-06-05	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 5/5